Aman Patel

SOFTWARE ENGINEER · APPLIED AI, BACKEND & DISTRIBUTED SYSTEMS

Greater Toronto Area, ON, Canada · Canadian Citizen · TN-eligible for US roles · Open to hybrid / remote · (416) 828-2451 · pateljaman@outlook.com · linkedin.com/in/amanjpatel · github.com/aman9824 · amanpatel.vercel.app

SUMMARY

Software engineer with **five years** building and operating production systems for **Fortune-50 enterprises**, now building production GenAI on that foundation. Since January 2025: designing, deploying, and operating a **production multi-agent system on Claude and Vertex AI** — agent orchestration (Claude Agent SDK, LangGraph), hybrid retrieval, tool design, and test-case-driven evaluation — for a legacy-platform modernization programme whose output is **Java 17 Spring Boot services**. Underneath that, four years of Java/Spring Boot and Python microservices on AWS, GCP, and OpenShift/Kubernetes, event-driven over Kafka, including **just over two years owning production incident triage on a 24/7 on-call rotation**: concurrency and race conditions, lock contention, and database index performance under live load. Works **customer-embedded end to end** — requirements discovery, technical specification, architecture, deployment, and ongoing support. B.S. Computer Science, York University.

CORE SKILLS

Languages: Python, **Java (JDK 17)**, TypeScript / JavaScript, C++, SQL, Bash

AI / LLM Engineering: Claude (Claude Agent SDK, Claude Code, Messages API, Vertex AI, Bedrock), **multi-agent orchestration (LangGraph, CrewAI)**, LangChain, **RAG** — hybrid lexical + dense retrieval, chunking, query rewriting; **MCP servers**, tool design, prompt engineering, **evaluation harnesses**, Gemini

Backend & Distributed Systems: Core Java, object-oriented design, **concurrency and multi-threading**, **Spring Boot**, Spring Security, Hibernate, FastAPI, Node.js, **microservices**, **REST API design**, event-driven and publish/subscribe (**Kafka**), high availability, caching, database index and query tuning, system design

Cloud, Containers & DevOps: GCP (Vertex AI), **AWS (Bedrock, EC2, S3, IAM)**, **Red Hat OpenShift**, **Kubernetes**, Docker and Docker Compose, Helm, Linux / Unix shell, **CI/CD (Jenkins, GitHub Actions)**, Git / GitHub / Bitbucket

Data: PostgreSQL, Oracle, Couchbase, **Elasticsearch**, Redis, vector stores

Front End: React, micro-frontends, TypeScript, HTML / CSS

Testing & Observability: JUnit, pytest, Selenium, mypy / ruff, Splunk, Dynatrace, Kibana

Ways of Working: Customer-embedded delivery, requirements discovery, technical specification, Agile, peer code review, mentoring, Jira / Confluence

Domains: Automotive, Telecommunications, Financial Services

EXPERIENCE

Software Engineer — GenAI (Contract)

Jan 2025 – Present

Cascade Cloud Tech · *Toronto, ON (Remote)*

Customer: a Fortune-50 automotive manufacturer — legacy-platform modernization program

- Designed and built a **production multi-agent system** (Python, Claude Agent SDK, LangGraph) with structured retrieval that converts legacy mainframe COBOL/JCL into **Java 17 Spring Boot services**. On the pilot application, delivery fell from **roughly six months on the prior all-manual baseline to roughly six weeks** with the GenAI-plus-developer workflow (measured on one application).
- **Own the target Java architecture** the generated output has to conform to — service boundaries, interfaces, and coding standards — alongside the pipeline that produces it.
- Built the retrieval layer: **hybrid lexical and dense search over Elasticsearch** with chunking and query rewriting, so generated code is grounded in the customer's own source and standards rather than model priors.
- **Deployed and operate the system on Vertex AI and OpenShift** as distributed services on Kubernetes — model invocation, retrieval, error handling, retries, and structured logging.
- Embedded with the customer's engineering teams **from discovery through production**: refined ambiguous business and system requirements into written technical specifications, and drove output accuracy through **test-case-driven evaluation**.

Software Developer (Freelance)

May 2024 – Dec 2024

16597629 Canada Inc. · *Toronto, ON*

- Built cloud-native backend services in **Java 17, Spring Boot, Spring Security, and Hibernate** on OpenShift and GCP for an enterprise modernization program.
- Designed **microservices around publish/subscribe messaging** for real-time data flow and high availability; configured OpenShift Deployments, Services, and Routes.
- Tuned application performance through **multi-threading and memory management**; automated operational tasks with Unix shell scripting.
- Instrumented services with Splunk and Dynatrace for log management and performance monitoring; ran code reviews and **mentored junior developers**.

Software Developer

Jan 2022 – Apr 2024

Amdocs · *Mississauga, ON*

Customer: a Fortune-50 US telecom carrier — 24/7 production systems serving millions of subscribers

- Spent **just over two years on the 24/7 on-call rotation** owning production incident triage for high-throughput Java/Spring services on AWS and OpenShift — diagnosing **concurrency and race conditions, lock contention, and database index performance** under live load.
- Built and maintained **React micro-frontends over Node.js and Spring Boot microservices**, end to end from analysis and design through implementation and support.
- Documented system components and architecture designs against **security, integration, and operability standards**; tracked analysis and delivery in Jira and Confluence.
- Embedded with the carrier's engineering and operations teams **across three countries** — translated end-user requirements into technical specifications, ran peer code review, and handed over open incidents cleanly at shift boundaries.

Quality Automation Engineer

Jul 2021 – Jan 2022

Equifax · *Toronto, ON*

Credit-risk scoring platform — regulated consumer financial data

- Built **Selenium and JUnit automation into the CI/CD/CT pipeline**, covering functional and non-functional behaviour (performance, reliability, security).
- Designed and maintained manual and automated test suites with **traceability matrices** derived from business requirements, design specifications, and architecture blueprints; supported defect resolution.

Software Developer Intern

Jan 2021 – Jun 2021

The Graduate Network · *Toronto, ON*

- Built full-stack web-application features as clean, testable code; translated user needs into technical specifications and worked to senior developers' code review and direction (IntelliJ, Kibana).

Microsoft Learn Student Ambassador

Jan 2021 – May 2021

Microsoft · *Ontario, Canada*

- Selected as **one of approximately 40 ambassadors across Canada**; ran live Azure enablement sessions and gathered requirements from university audiences.

SELECTED PERSONAL PROJECTS (OPEN SOURCE)

AI Filings Analyst — multi-agent extraction with verified citations. Claude system that pulls KPIs and risk signals out of 10-K/10-Q filings with **span-level citations**. Hybrid BM25 and dense retrieval; a **citation verifier that rejects any claim the source span does not support**; evaluation thresholds enforced in CI. github.com/aman9824/finance-filings-analyst

YTC — dual-engine research platform. Separates a slow, model-driven analysis path from a fast, fully deterministic execution engine gated by arithmetic checks — **no model in the hot path**. **C++ gate exposed via pybind11** and parity-tested against the Python reference; asyncio pipeline; a backtester that replays the exact live decision path rather than a reimplementation of it; FastAPI dashboard. github.com/aman9824/ytc

CERTIFICATIONS

AWS Certified Cloud Practitioner · Become a Spring Boot Developer · Kubernetes: Package Management with Helm · Docker for Developers · Learn Apache Kafka for Beginners

Anthropic Academy (2026): Building with the Claude API · MCP (introductory and advanced) · Sub-agents · Agent Skills · Claude Code · Vertex AI · Bedrock

EDUCATION

York University · B.S. in Computer Science

Jan 2019 – Jun 2021

Toronto, ON · accelerated programme, completed without summer breaks